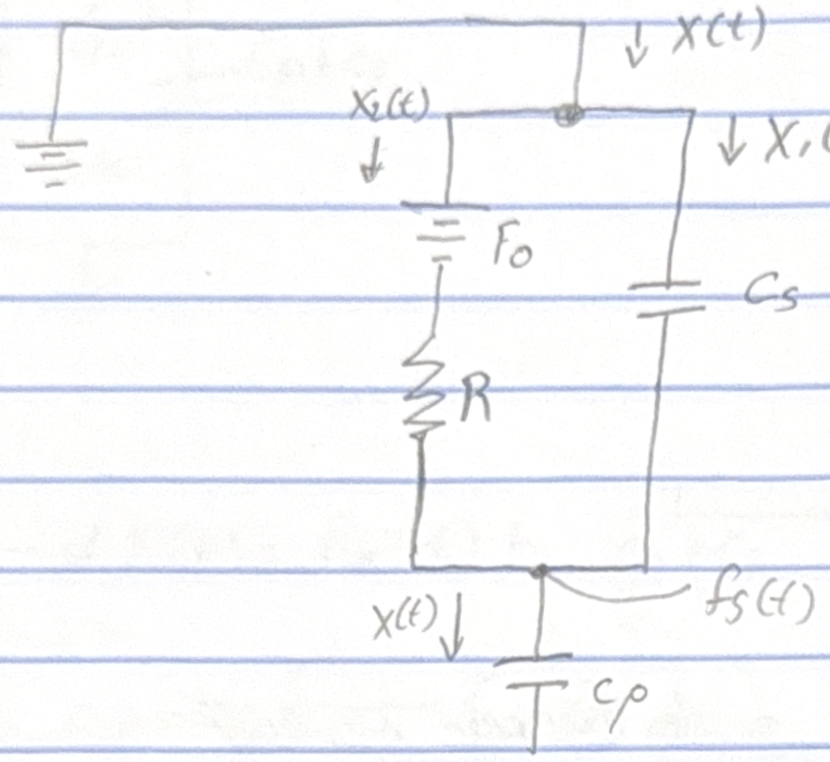
26/oct/26

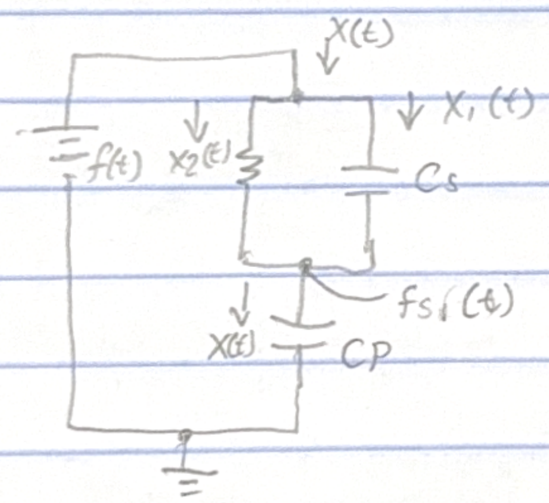
Practica #3 (5.5) Modelado y control del sistema musculoesqueletico

Transformación de sistema mecanico a electrico.

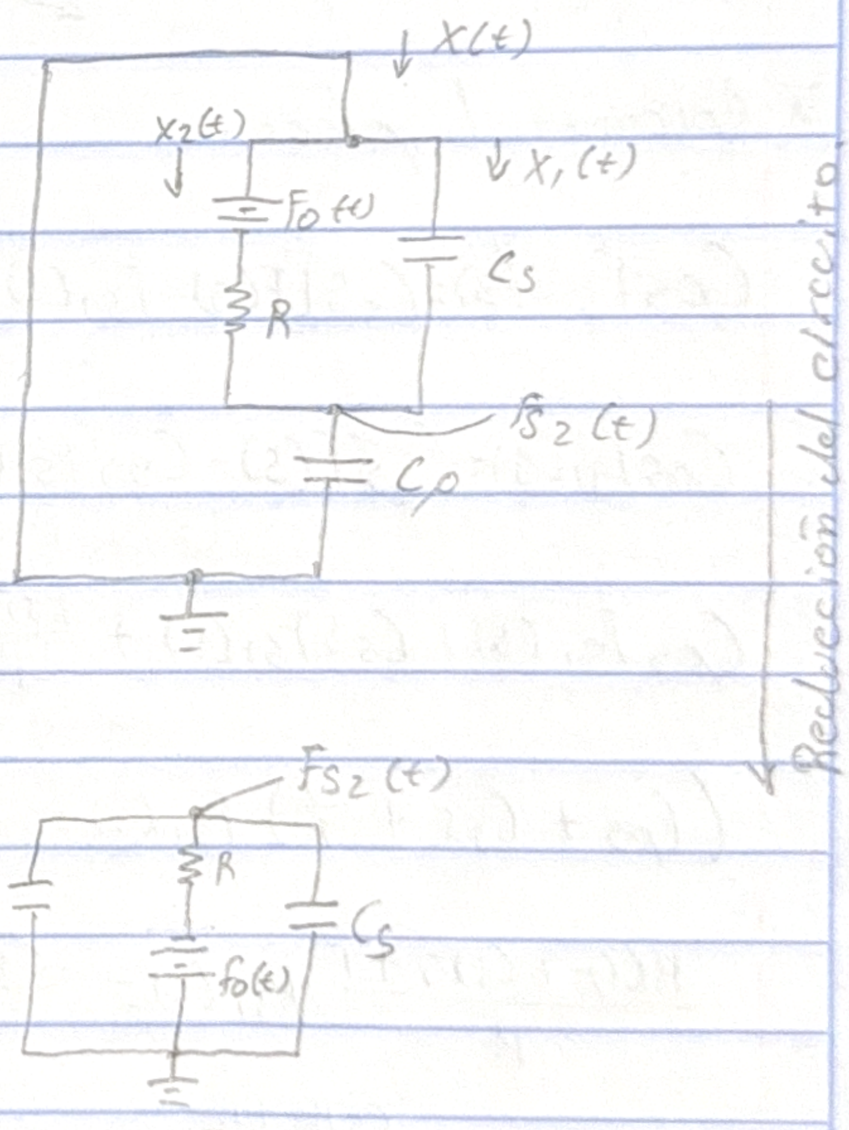


$F_S(t) =$

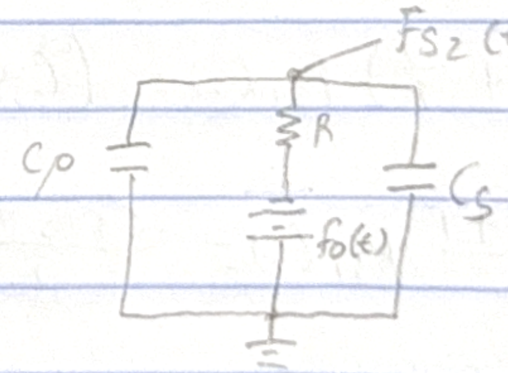
* Superposición ($F_{S1}(t)$)

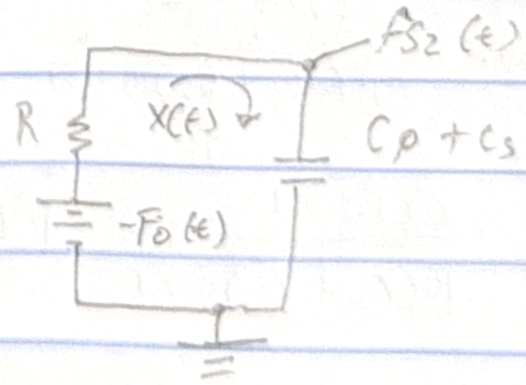


* Superposicion ($F_{S2}(t)$)



Reducción del circuito





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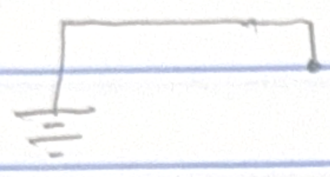
* Analisis matematico.
Debido a que el sistema contiene dos fuentes de voltaje, se debe aplicar el principio de superposición.

- Circuito #1 / apagando $F_0(t)$



$X(t) = X_1(t) + X_2(t)$

$X(t) = C_p \frac{dF_{S1}(t)}{dt}$

$X_1(t) = C_S \frac{d[F(t) - F_{S1}(t)]}{dt}$

$X_2(t) = \frac{F(t) - F_{S1}(t)}{R}$

* Sustituimos en la función principal

$C_p \frac{dF_{S1}(t)}{dt} = C_S \frac{d[F(t) - F_{S1}(t)]}{dt} + \frac{F(t) - F_{S1}(t)}{R}$

* Aplicamos Laplace:

$C_p S F_{S1}(S) = C_S S [F(S) - F_{S1}(S)] + \frac{F(S) - F_{S1}(S)}{R}$

$C_p S F_{S1}(S) = C_S S F(S) - C_S S F_{S1}(S) + \frac{F(S)}{R} - \frac{F_{S1}(S)}{R}$

$C_p S F_{S1}(S) + C_S S F_{S1}(S) + \frac{F_{S1}(S)}{R} = C_S S F(S) + \frac{F(S)}{R}$

$\left(C_p S + C_S S + \frac{1}{R}\right) F_{S1}(S) = \left(C_S S + \frac{1}{R}\right) F(S)$

$\frac{R(C_p + C_S)S + 1}{R} F_{S1}(S) = \frac{C_S R S + 1}{R} F(S)$

$F_{S1}(S) = \frac{\frac{C_S R S + 1}{R}}{\frac{R(C_p + C_S)S + 1}{R}} F(S)$

$F_{S1}(S) = \frac{C_S R S + 1}{R(C_p + C_S)S + 1} F(S)$

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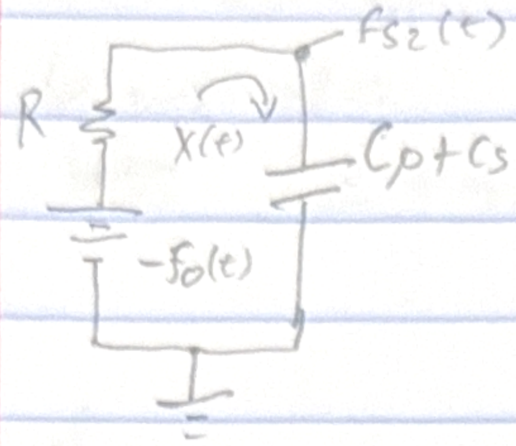
* Circuito #2: Apagando $F(t)$

Recordando que $F_0(t) \propto F(t)$



* Analisis:

$$-\alpha F(t) = R x(t) + \frac{1}{C_p+C_s}\int x(t)\,dt$$

$$F_{S2}(t) = \frac{1}{C_p+C_s}\int x(t)\,dt$$

* Laplace:

$$-\alpha F(s) = R x(s) + \frac{x(s)}{(C_p+C_s)s}$$

$$F_{S2}(s) = \frac{x(s)}{(C_p+C_s)s}$$

* Función de Transfer. Segundo Circuito: $\frac{F_{S2}(t)}{-\alpha F(s)}$

$$F(s) = -\frac{1}{\alpha}\left(R + \frac{1}{(C_p+C_s)s}\right)X(s) = \frac{1}{\alpha}\cdot\frac{R(C_p+C_s)s+1}{(C_p+C_s)s}X(s)$$

$$\frac{F_{S2}(s)}{F(t)} = \frac{\frac{x(s)}{(C_p+C_s)s}}{-\frac{1}{\alpha}\frac{R(C_p+C_s)s+1}{(C_p+C_s)s}X(s)} = -\frac{\alpha F(t)}{R(C_p+C_s)s+1}$$

* Función de transferencia:

$$F_S(s) = F_{S1}(s) + F_{S2}(s) = \frac{C_s R s+1}{R(C_p+C_s)s+1}F(s) - \frac{\alpha F(s)}{R(C_p+C_s)s+1}$$

$$F_S(s) = \frac{C_s R s+1-\alpha}{R(C_p+C_s)s+1}F(s) \quad \therefore \quad \frac{F_S(s)}{F(s)} = \frac{C_s R s+1-\alpha}{R(C_p+C_s)s+1}$$

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* Error en estado estacionario:

$$e(s) = \lim_{s \to \emptyset} s F(s) \left(1 - \frac{F_8(s)}{F(s)}\right) = s \frac{1}{s} \left(1 - \frac{RC_s s + 1 - \alpha}{(RC_p + RC_s)s + 1}\right)$$

$$e(s) = 1 - \frac{1-\alpha}{1} = 1 - 1 + \alpha = \alpha$$

* Estabilidad en lazo abierto:

$$(RC_p + RC_s)s + 1 = \emptyset$$

$\lambda = -\frac{1}{RC_p + RC_s}$ = Sistema estable. (Numero real negativo)

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